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Course: Network Forensics
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Abstract

This lab focuses on analyzing multiple packet capture (PCAP) files using command-line tools to identify protocols, authentication activity, and file transfer behavior. Tools such as shasum, capinfos, tshark, and grep were used to verify file integrity, extract metadata, summarize host/protocol activity, and isolate authentication and file transfer indicators.

Across the captures, unencrypted FTP traffic exposed plaintext credentials and a file download request, demonstrating the risk of legacy protocols. HTTP traffic was also reviewed to determine whether downloads occurred and whether authentication attempts were successful. Overall, this exercise demonstrates how repeatable command-line packet analysis supports incident response and evidence-driven conclusions.

Introduction

The purpose of this assignment is to analyze packet capture files from systems suspected of compromise and determine what network activity occurred. Specifically, the analysis aims to (1) verify file integrity, (2) identify active hosts and protocols, (3) determine whether authentication attempts occurred, and (4) confirm whether files were downloaded or uploaded using FTP or HTTP.

Tools and Commands Used

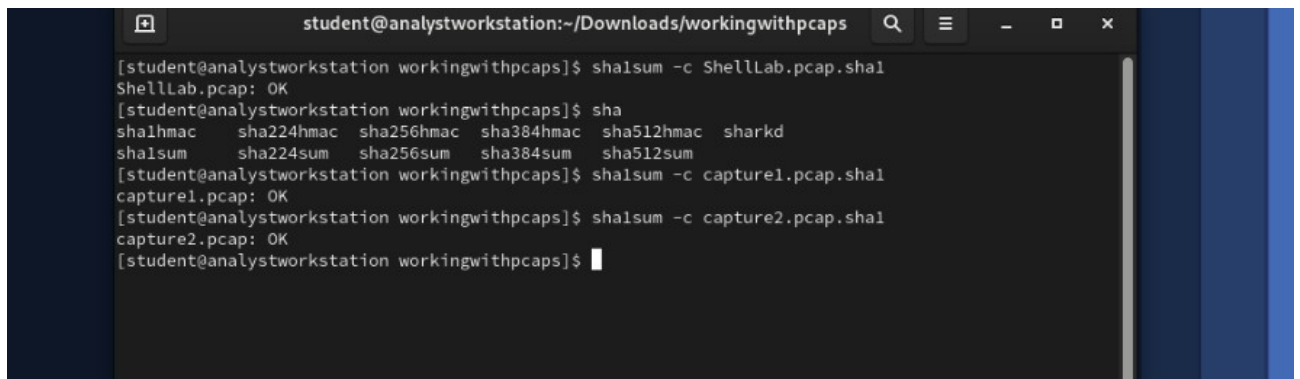
- shasum -c <file>.sha1 (integrity verification)
- capinfos <file>.pcap (capture metadata)
- tshark -r <file>.pcap -q -z ip_hosts,tree (host/IP statistics)
- tshark -r <file>.pcap -q -z io,phs (protocol hierarchy statistics)
- tshark display filters (e.g., ftp, http, ftp.request.command == RETR)
- grep for indicators (e.g., USER/PASS, GET/POST, PUT)

Analysis and Findings

1. ShellLab.pcap Results

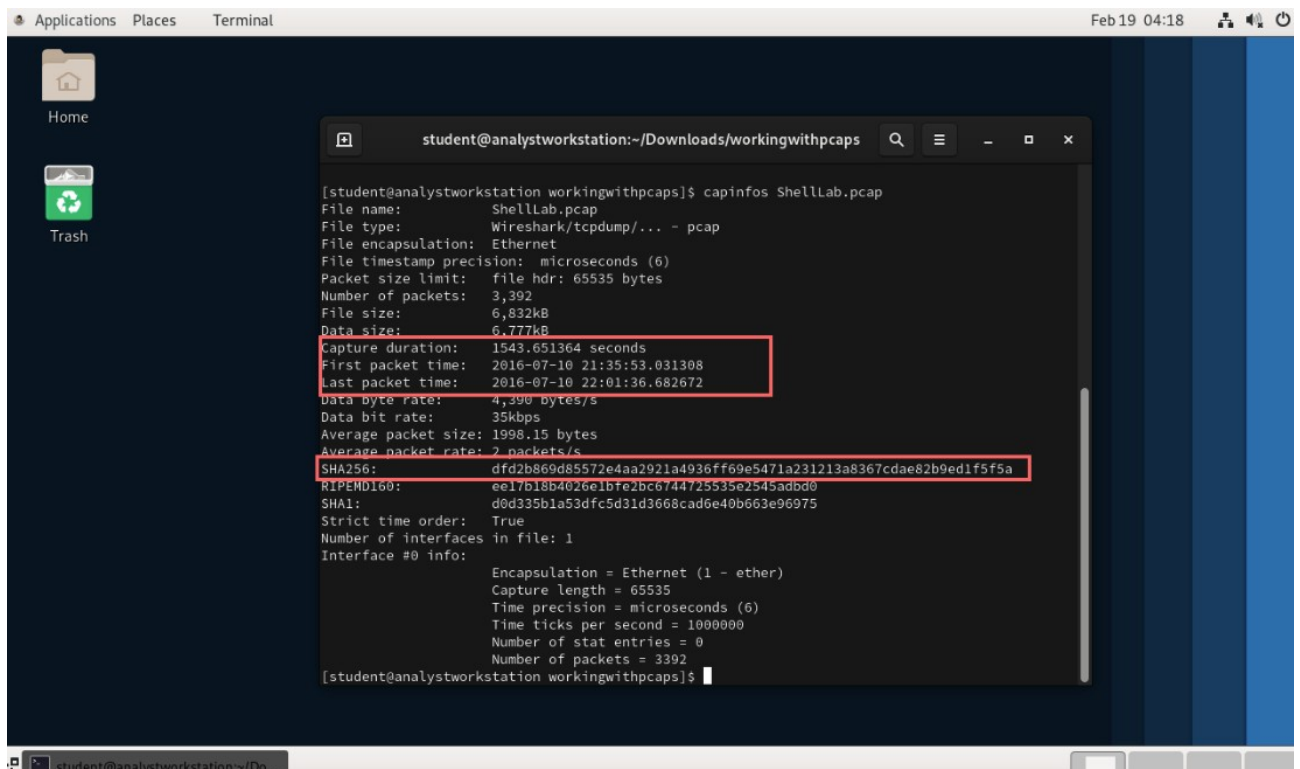
1.1 File Integrity and Metadata

The SHA1 verification for ShellLab.pcap returned OK, confirming the capture file was not modified after creation. Using capinfos, the capture duration and timestamp range were recorded for scoping the analysis window.



```
student@analystworkstation:~/Downloads/workingwithpcaps
[student@analystworkstation workingwithpcaps]$ sha1sum -c ShellLab.pcap.sha1
ShellLab.pcap: OK
[student@analystworkstation workingwithpcaps]$ sha
sha1hmac  sha224hmac  sha256hmac  sha384hmac  sha512hmac  sharkd
sha1sum    sha224sum  sha256sum  sha384sum  sha512sum
[student@analystworkstation workingwithpcaps]$ sha1sum -c capture1.pcap.sha1
capture1.pcap: OK
[student@analystworkstation workingwithpcaps]$ sha1sum -c capture2.pcap.sha1
capture2.pcap: OK
[student@analystworkstation workingwithpcaps]$
```

Figure 1.1. SHA1 verification for ShellLab.pcap showing OK status for file integrity.



```
Applications  Places  Terminal  Feb 19 04:18
student@analystworkstation:~/Downloads/workingwithpcaps
[student@analystworkstation workingwithpcaps]$ capinfos ShellLab.pcap
File name:          ShellLab.pcap
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  3,392
File size:          6,832kB
Data size:          6,777kB
Capture duration:   1543.651364 seconds
First packet time:  2016-07-10 21:35:53.031308
Last packet time:   2016-07-10 22:01:36.682672
Data byte rate:     4,390 bytes/s
Data bit rate:      35kbps
Average packet size: 1998.15 bytes
Average packet rate: 2 packets/s
SHA256:             dfd2b869d85572e4aa2921a4936ff69e5471a231213a8367cdae82b9ed1f5f5a
RIPEMD160:          ee17b18b4026e1b7e2bc6744725535e2545adb0
SHA1:               d0d335b1a53dfc5d31d3668cad6e40b663e96975
Strict time order:  True
Number of interfaces in file: 1
Interface #0 info:
  Encapsulation = Ethernet (1 - ether)
  Capture length = 65535
  Time precision = microseconds (6)
  Time ticks per second = 1000000
  Number of stat entries = 0
  Number of packets = 3392
[student@analystworkstation workingwithpcaps]$
```

Figure 1.2. *capinfos* output for *ShellLab.pcap* showing capture duration and timestamp range.

1.2 Host and Protocol Identification

IPv4 host statistics show the dominant internal host was 172.31.21.16 with the highest packet count. Protocol hierarchy statistics indicate a mix of DNS, HTTP, TLS, SMTP, and FTP/FTP-DATA traffic, with FTP present as a notable plaintext service.

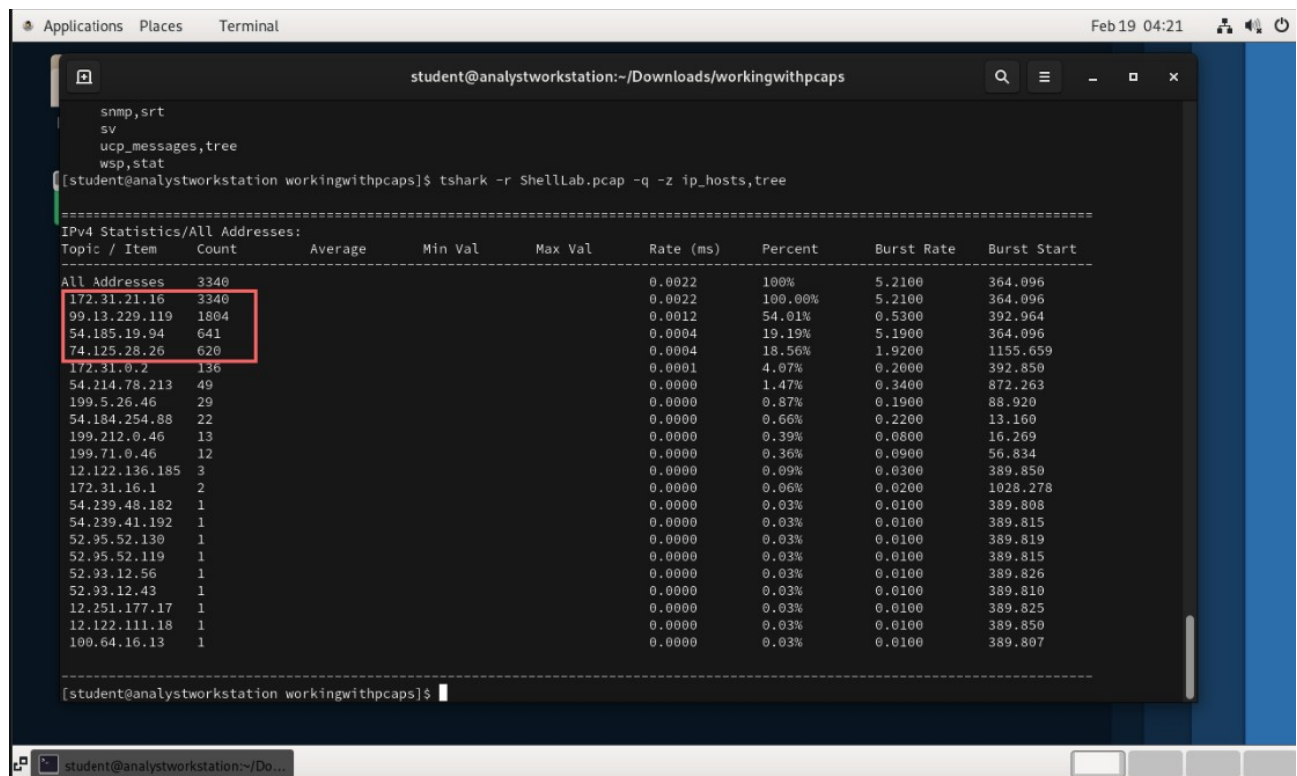


Figure 1.3. Host/IP statistics (*ip_hosts,tree*) for *ShellLab.pcap* highlighting top talkers.

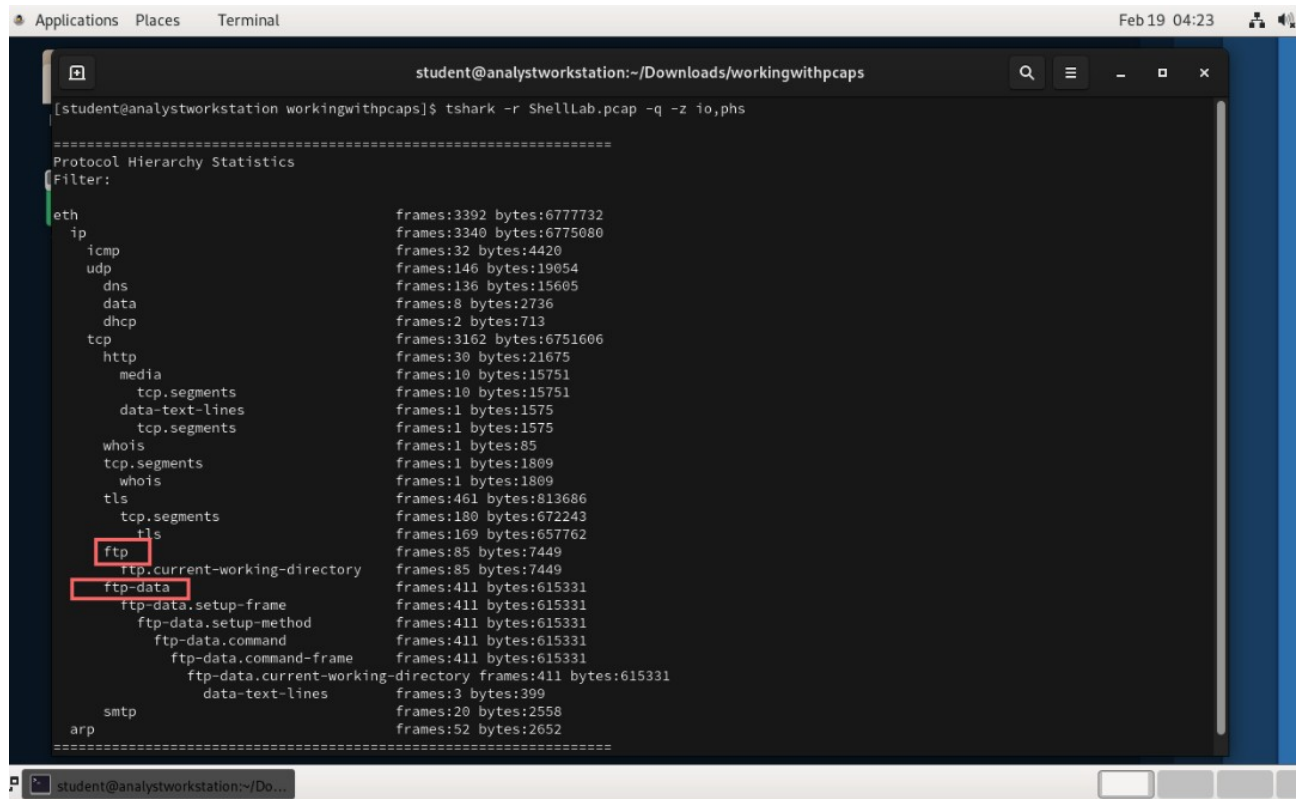


Figure 1.4. Protocol hierarchy statistics (io,phs) for ShellLab.pcap highlighting FTP and FTP-DATA.

1.3 FTP Activity and File Transfer Indicators

FTP traffic was isolated and written to a separate pcap for focused analysis. A RETR command was observed for the file xfil.rar, indicating a file download from the FTP server. No STOR commands were observed, suggesting no file uploads occurred over FTP during the capture.

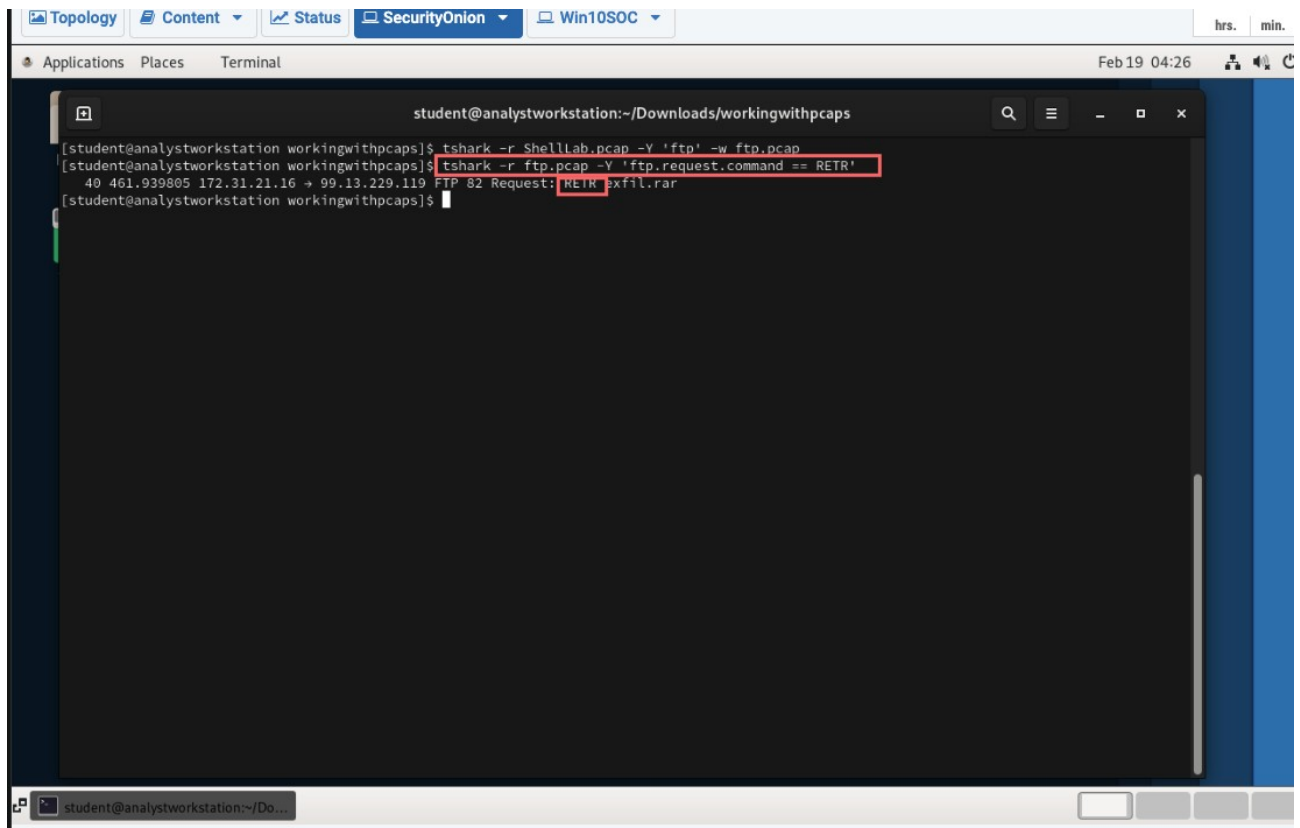


Figure 1.5. Filtering ShellLab.pcap for FTP traffic and identifying a RETR request for xfil.rar.

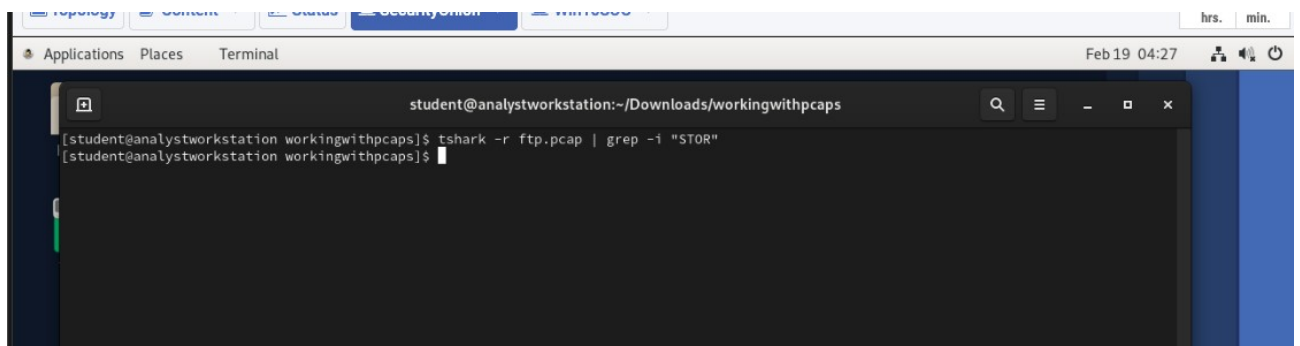


Figure 1.6. Searching for FTP STOR commands returned no results, indicating no FTP uploads.

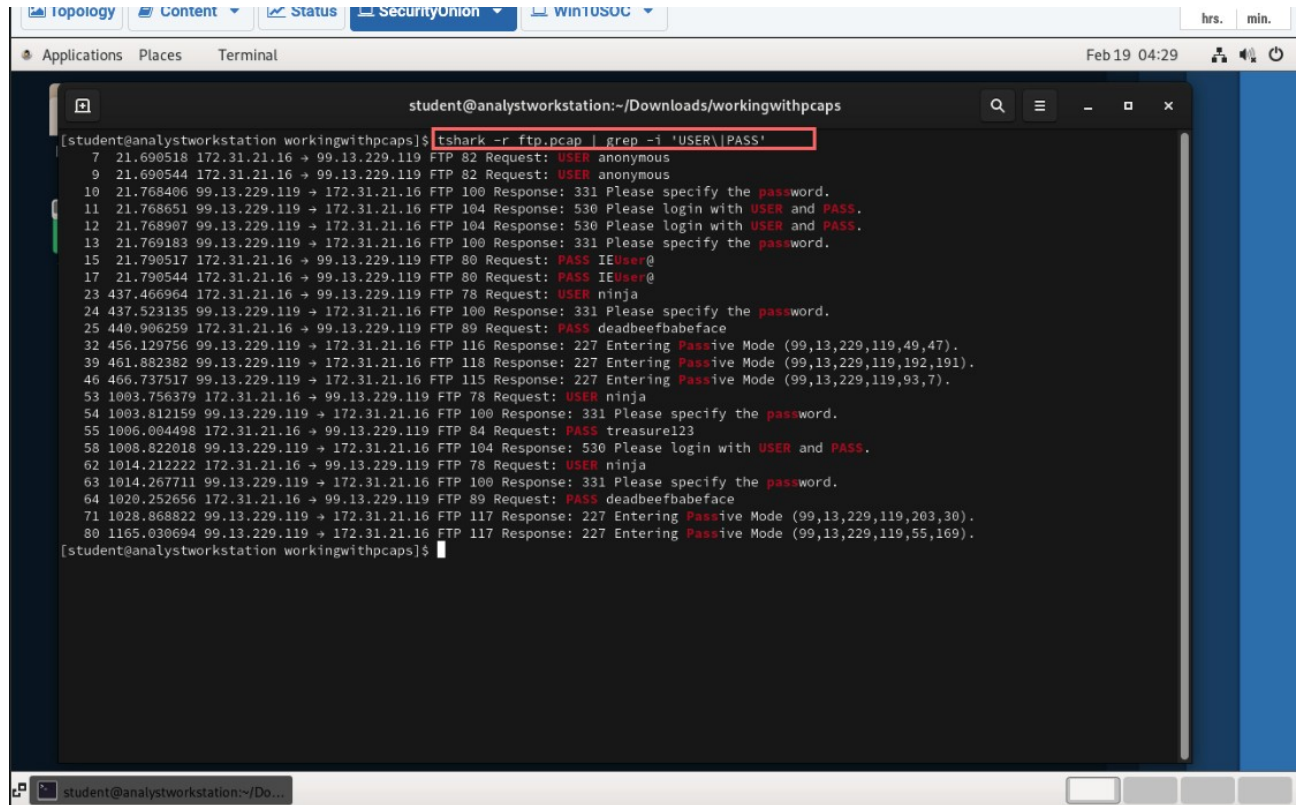
1.4 FTP Authentication Evidence

Filtering for USER and PASS commands revealed plaintext credentials transmitted in the FTP session. Both failed and successful login attempts were present. Successful authentication was identified for:

- Username: ninja

- Password: deadbeefbabeface

A failed login attempt was also observed for ninja using the password treasure123.



```
student@analystworkstation:~/Downloads/workingwithpcaps
[student@analystworkstation workingwithpcaps]$ tshark -r ftp.pcap | grep -i 'USER\\|PASS'
 7 21.690518 172.31.21.16 -> 99.13.229.119 FTP 82 Request: USER anonymous
 9 21.690544 172.31.21.16 -> 99.13.229.119 FTP 82 Request: USER anonymous
10 21.768406 99.13.229.119 -> 172.31.21.16 FTP 100 Response: 331 Please specify the password.
11 21.768651 99.13.229.119 -> 172.31.21.16 FTP 104 Response: 530 Please login with USER and PASS.
12 21.768907 99.13.229.119 -> 172.31.21.16 FTP 104 Response: 530 Please login with USER and PASS.
13 21.769183 99.13.229.119 -> 172.31.21.16 FTP 100 Response: 331 Please specify the password.
15 21.790517 172.31.21.16 -> 99.13.229.119 FTP 80 Request: PASS IEUser@
17 21.790544 172.31.21.16 -> 99.13.229.119 FTP 80 Request: PASS IEUser@
23 437.466964 172.31.21.16 -> 99.13.229.119 FTP 78 Request: USER ninja
24 437.523135 99.13.229.119 -> 172.31.21.16 FTP 100 Response: 331 Please specify the password.
25 440.906259 172.31.21.16 -> 99.13.229.119 FTP 89 Request: PASS deadbeefbabeface
32 456.129756 99.13.229.119 -> 172.31.21.16 FTP 116 Response: 227 Entering Passive Mode (99,13,229,119,49,47).
39 461.882382 99.13.229.119 -> 172.31.21.16 FTP 118 Response: 227 Entering Passive Mode (99,13,229,119,192,191).
46 466.737517 99.13.229.119 -> 172.31.21.16 FTP 115 Response: 227 Entering Passive Mode (99,13,229,119,93,7).
53 1003.756379 172.31.21.16 -> 99.13.229.119 FTP 78 Request: USER ninja
54 1003.812159 99.13.229.119 -> 172.31.21.16 FTP 100 Response: 331 Please specify the password.
55 1006.004498 172.31.21.16 -> 99.13.229.119 FTP 84 Request: PASS treasure123
58 1008.822018 99.13.229.119 -> 172.31.21.16 FTP 104 Response: 530 Please login with USER and PASS.
62 1014.212222 172.31.21.16 -> 99.13.229.119 FTP 78 Request: USER ninja
63 1014.267711 99.13.229.119 -> 172.31.21.16 FTP 100 Response: 331 Please specify the password.
64 1020.252656 172.31.21.16 -> 99.13.229.119 FTP 89 Request: PASS deadbeefbabeface
71 1028.868822 99.13.229.119 -> 172.31.21.16 FTP 117 Response: 227 Entering Passive Mode (99,13,229,119,203,30).
80 1165.030694 99.13.229.119 -> 172.31.21.16 FTP 117 Response: 227 Entering Passive Mode (99,13,229,119,55,169).
[student@analystworkstation workingwithpcaps]$
```

Figure 1.7. Plaintext FTP authentication attempts (USER/PASS) extracted from ftp.pcap.

1.5 HTTP Request Review

HTTP GET requests were observed to multiple hosts, including package repositories and web resources. A search for HTTP PUT activity returned no evidence of file uploads within the provided output, supporting the conclusion that HTTP traffic in this capture primarily consisted of downloads/requests rather than uploads.


```
Topology Content Status Security Onion winUSOC hrs. min.
Applications Places Terminal Feb 19 04:43

student@analystworkstation:~/Downloads/workingwithpcaps

http://us-west-2.ec2.archive.ubuntu.com/ubuntu/pool/main/libl/liblinear/liblinear-tools_1.8%2bdfsg-1ubuntu1_amd64.deb
http://us-west-2.ec2.archive.ubuntu.com/ubuntu/pool/main/n/nmap/nmap_6.40-0.2ubuntu1_amd64.deb
http://99-13-229-119.lightspeed.snjcsa.sbcglobal.net/.git/HEAD
http://99-13-229-119.lightspeed.snjcsa.sbcglobal.net/
http://99-13-229-119.lightspeed.snjcsa.sbcglobal.net/robots.txt
http://99-13-229-119.lightspeed.snjcsa.sbcglobal.net/
http://99-13-229-119.lightspeed.snjcsa.sbcglobal.net/
http://99-13-229-119.lightspeed.snjcsa.sbcglobal.net/favicon.ico
http://us-west-2.ec2.archive.ubuntu.com/ubuntu/pool/main/libs/libsocket6-perl/libsocket6-perl_0.25-1_amd64.deb
http://us-west-2.ec2.archive.ubuntu.com/ubuntu/pool/main/libi/libio-socket-inet6-perl/libio-socket-inet6-perl_2.71-1_all.deb
http://us-west-2.ec2.archive.ubuntu.com/ubuntu/pool/universe/s/sendmail/sendmail_1.56-5_all.deb
http://us-west-2.ec2.archive.ubuntu.com/ubuntu/pool/multiverse/r/rar/rar_4.2.0-1_amd64.deb
[student@analystworkstation workingwithpcaps]$ tshark -r ShellLab.pcap | gre -i "GET"
bash: gre: command not found
[student@analystworkstation workingwithpcaps]$ tshark -r ShellLab.pcap | grep -i "GET"
14 13.160593 172.31.21.16 + 54.184.254.88 HTTP 218 GET /ubuntu/pool/main/w/whois/whois_5.1.1_amd64.deb HTTP/1.1
57 56.833603 172.31.21.16 + 199.71.0.46 HTTP 153 GET /rest/nets HTTP/1.1
122 364.096601 172.31.21.16 + 54.185.19.94 HTTP 227 GET /ubuntu/pool/main/l/luas5.2/libluas5.2-0_5.2.3-1_amd64.deb HTTP/1.1
155 364.100992 172.31.21.16 + 54.185.19.94 HTTP 229 GET /ubuntu/pool/main/b/bias/libblas3_1.2.20110419-7_amd64.deb HTTP/1.1
211 364.107583 172.31.21.16 + 54.185.19.94 HTTP 244 GET /ubuntu/pool/main/libl/liblinear/liblinear1_1.8%2bdfsg-1ubuntu1_amd64.deb HTTP/1.1
227 364.109795 172.31.21.16 + 54.185.19.94 HTTP 249 GET /ubuntu/pool/main/libl/liblinear/liblinear-tools_1.8%2bdfsg-1ubuntu1_amd64.deb HTTP/1.1
236 364.111741 172.31.21.16 + 54.185.19.94 HTTP 226 GET /ubuntu/pool/main/n/nmap/nmap_6.40-0.2ubuntu1_amd64.deb HTTP/1.1
1144 393.801591 172.31.21.16 + 99.13.229.119 HTTP 257 GET /.git/HEAD HTTP/1.1
1172 393.900546 172.31.21.16 + 99.13.229.119 HTTP 248 GET / HTTP/1.1
1173 393.900573 172.31.21.16 + 99.13.229.119 HTTP 258 GET /robots.txt HTTP/1.1
1366 402.345615 172.31.21.16 + 99.13.229.119 HTTP 248 GET / HTTP/1.1
1390 410.728065 172.31.21.16 + 99.13.229.119 HTTP 248 GET / HTTP/1.1
1416 419.108181 172.31.21.16 + 99.13.229.119 HTTP 248 GET / HTTP/1.1
1443 427.490634 172.31.21.16 + 99.13.229.119 HTTP 259 GET /favicon.ico HTTP/1.1
2412 872.264330 172.31.21.16 + 54.214.78.213 HTTP 242 GET /ubuntu/pool/main/libs/libsocket6-perl/libsocket6-perl_0.25-1_amd64.deb HTTP/1.1
2430 872.269972 172.31.21.16 + 54.214.78.213 HTTP 256 GET /ubuntu/pool/main/libi/libio-socket-inet6-perl/libio-socket-inet6-perl_2.71-1_all.deb HTTP/1.1
2441 872.273036 172.31.21.16 + 54.214.78.213 HTTP 229 GET /ubuntu/pool/universe/s/sendmail/sendmail_1.56-5_all.deb HTTP/1.1
2543 1076.319125 172.31.21.16 + 54.185.19.94 HTTP 222 GET /ubuntu/pool/multiverse/r/rar/rar_4.2.0-1_amd64.deb HTTP/1.1
[student@analystworkstation workingwithpcaps]$
```

Figure 1.8. HTTP GET requests extracted from ShellLab.pcap using tshark and grep.

```
student@analystworkstation:~/Downloads/workingwithpcaps

[student@analystworkstation workingwithpcaps]$ tshark -r ShellLab.pcap | grep -i "PUT"
105 248.688882 172.31.21.16 + 172.31.0.2 DNS 102 Standard query 0xca06 AAAA ip-172-31-21-16.us-west-2.compute.internal
106 248.682212 172.31.0.2 + 172.31.21.16 DNS 163 Standard query response 0xca06 AAAA ip-172-31-21-16.us-west-2.compute.internal SOA ns0.us-west-2.c
ompute.internal
109 248.700506 172.31.21.16 + 172.31.0.2 DNS 102 Standard query 0x269f A ip-172-31-21-16.us-west-2.compute.internal
110 248.701745 172.31.0.2 + 172.31.21.16 DNS 118 Standard query response 0x269f A ip-172-31-21-16.us-west-2.compute.internal A 172.31.21.16
2480 962.887364 172.31.21.16 + 172.31.0.2 DNS 102 Standard query 0x9fc5 A ip-172-31-21-16.us-west-2.compute.internal
2481 962.887396 172.31.21.16 + 172.31.0.2 DNS 102 Standard query 0xd89e AAAA ip-172-31-21-16.us-west-2.compute.internal
2482 962.888610 172.31.0.2 + 172.31.21.16 DNS 118 Standard query response 0x9fc5 A ip-172-31-21-16.us-west-2.compute.internal A 172.31.21.16
2483 962.893057 172.31.0.2 + 172.31.21.16 DNS 163 Standard query response 0xd89e AAAA ip-172-31-21-16.us-west-2.compute.internal SOA ns0.us-west-2.c
ompute.internal
2486 1012.728437 172.31.21.16 + 172.31.0.2 DNS 102 Standard query 0xe0df A ip-172-31-21-16.us-west-2.compute.internal
2487 1012.729866 172.31.0.2 + 172.31.21.16 DNS 118 Standard query response 0xe0df A ip-172-31-21-16.us-west-2.compute.internal A 172.31.21.16
2500 1012.804759 172.31.0.2 + 172.31.21.16 DNS 141 Standard query response 0x816d PTR 16.21.31.172.in-addr.arpa PTR ip-172-31-21-16.us-west-2.compu
e.internal
2503 1012.839214 172.31.21.16 + 74.125.28.26 SMTP 115 C: EHLO ip-172-31-21-16.us-west-2.compute.internal
2505 1012.857158 74.125.28.26 + 172.31.21.16 SMTP 234 S: 250-mx.google.com at your service, [52.10.52.243] | SIZE 157286400 | 8BITIME | STARTTLS | EN
HANCEDSTATUSCODES | PIPELINING | CHUNKING | SMTPUTF8
2664 1136.114078 172.31.21.16 + 172.31.0.2 DNS 102 Standard query 0x85c7 A ip-172-31-21-16.us-west-2.compute.internal
2665 1136.116190 172.31.0.2 + 172.31.21.16 DNS 118 Standard query response 0x85c7 A ip-172-31-21-16.us-west-2.compute.internal A 172.31.21.16
2678 1136.196947 172.31.0.2 + 172.31.21.16 DNS 141 Standard query response 0xc747 PTR 16.21.31.172.in-addr.arpa PTR ip-172-31-21-16.us-west-2.compu
e.internal
2681 1136.230834 172.31.21.16 + 74.125.28.26 SMTP 115 C: EHLO ip-172-31-21-16.us-west-2.compute.internal
2683 1136.246607 74.125.28.26 + 172.31.21.16 SMTP 234 S: 250-mx.google.com at your service, [52.10.52.243] | SIZE 157286400 | 8BITIME | STARTTLS | EN
HANCEDSTATUSCODES | PIPELINING | CHUNKING | SMTPUTF8
2823 1155.401688 172.31.21.16 + 172.31.0.2 DNS 102 Standard query 0xf6b3 A ip-172-31-21-16.us-west-2.compute.internal
2824 1155.402243 172.31.0.2 + 172.31.21.16 DNS 118 Standard query response 0xf6b3 A ip-172-31-21-16.us-west-2.compute.internal A 172.31.21.16
2837 1155.497753 172.31.0.2 + 172.31.21.16 DNS 141 Standard query response 0x8c14 PTR 16.21.31.172.in-addr.arpa PTR ip-172-31-21-16.us-west-2.compu
e.internal
2840 1155.513533 172.31.21.16 + 74.125.28.26 SMTP 115 C: EHLO ip-172-31-21-16.us-west-2.compute.internal
2842 1155.531619 74.125.28.26 + 172.31.21.16 SMTP 234 S: 250-mx.google.com at your service, [52.10.52.243] | SIZE 157286400 | 8BITIME | STARTTLS | EN
HANCEDSTATUSCODES | PIPELINING | CHUNKING | SMTPUTF8
3055 1164.498432 172.31.21.16 + 172.31.0.2 DNS 102 Standard query 0x5637 A ip-172-31-21-16.us-west-2.compute.internal
3056 1164.499562 172.31.0.2 + 172.31.21.16 DNS 118 Standard query response 0x5637 A ip-172-31-21-16.us-west-2.compute.internal A 172.31.21.16
3074 1195.658492 172.31.0.2 + 172.31.21.16 DNS 141 Standard query response 0x5d19 PTR 16.21.31.172.in-addr.arpa PTR ip-172-31-21-16.us-west-2.compu
e.internal

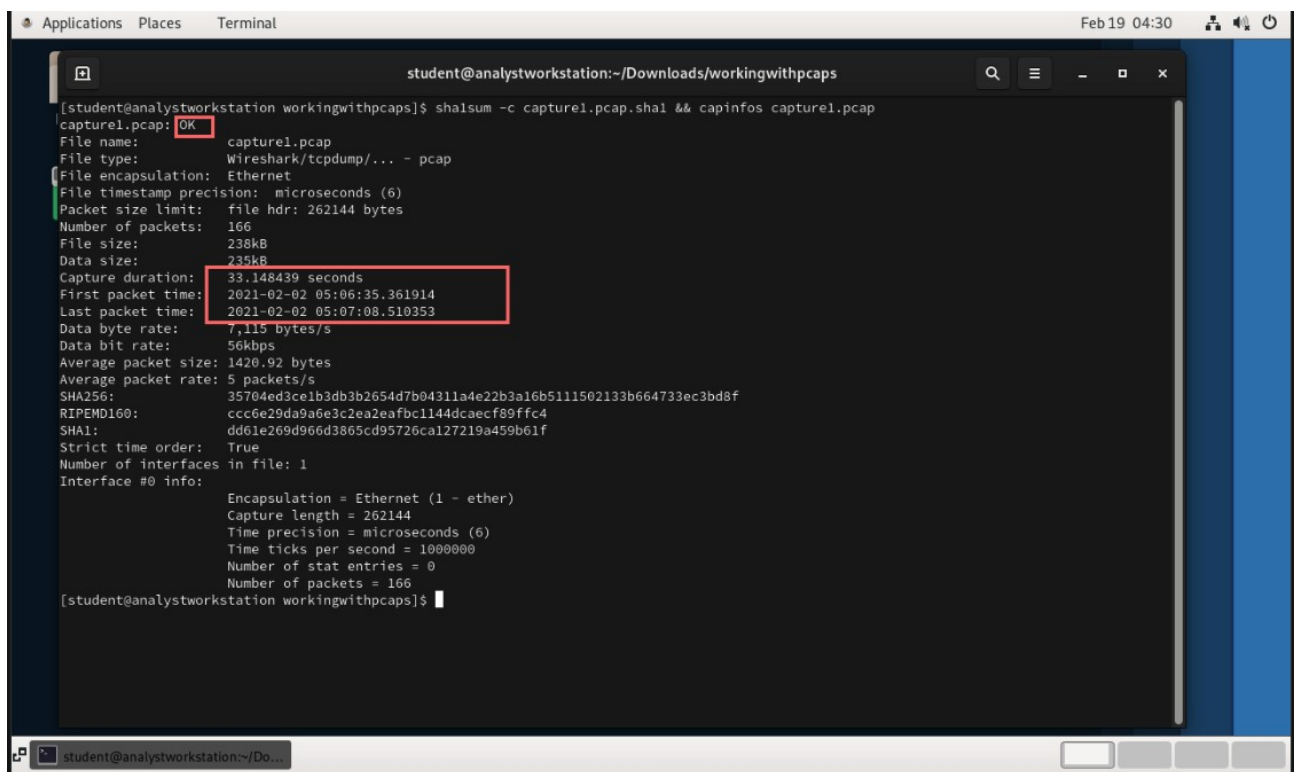
student@analystworkstation:~/Do...
```

Figure 1.9. Searching for HTTP PUT indicators; output shows no HTTP PUT uploads identified.

2. Capture1.pcap Results

2.1 File Integrity and Metadata

SHA1 verification for capture1.pcap returned OK. capinfos shows a short capture window (~33 seconds) with 166 packets, which limits the scope of observable activity but is sufficient to identify high-level protocols.



```
student@analystworkstation:~/Downloads/workingwithpcaps
[student@analystworkstation workingwithpcaps]$ shasum -c capture1.pcap.sha1 && capinfos capture1.pcap
capture1.pcap: OK
File name: capture1.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 262144 bytes
Number of packets: 166
File size: 238kB
Data size: 235kB
Capture duration: 33.148439 seconds
First packet time: 2021-02-02 05:06:35.361914
Last packet time: 2021-02-02 05:07:08.510353
Data byte rate: 7,115 bytes/s
Data bit rate: 56kbps
Average packet size: 1420.92 bytes
Average packet rate: 5 packets/s
SHA256: 35704ed3ce1b3db3b2654d7b04311a4e22b3a16b5111502133b664733ec3bd8f
RIPEMD160: ccc6e29da9a6e3c2ea2eafbc1144dcaecf89ffc4
SHA1: dd61e269d966d3865cd95726ca127219a459b61f
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
  Encapsulation = Ethernet (1 - ether)
  Capture length = 262144
  Time precision = microseconds (6)
  Time ticks per second = 1000000
  Number of stat entries = 0
  Number of packets = 166
[student@analystworkstation workingwithpcaps]$
```

Figure 2.1. SHA1 verification (OK) and capinfos metadata for capture1.pcap (packet count and time range).

2.2 Host and Protocol Identification

Host statistics show the capture is dominated by the internal host 172.31.0.2 communicating primarily with 102.129.202.163. Protocol hierarchy statistics indicate IP/TCP with limited HTTP activity and an image/jpeg artifact (image-jfif), suggesting web-based content retrieval.

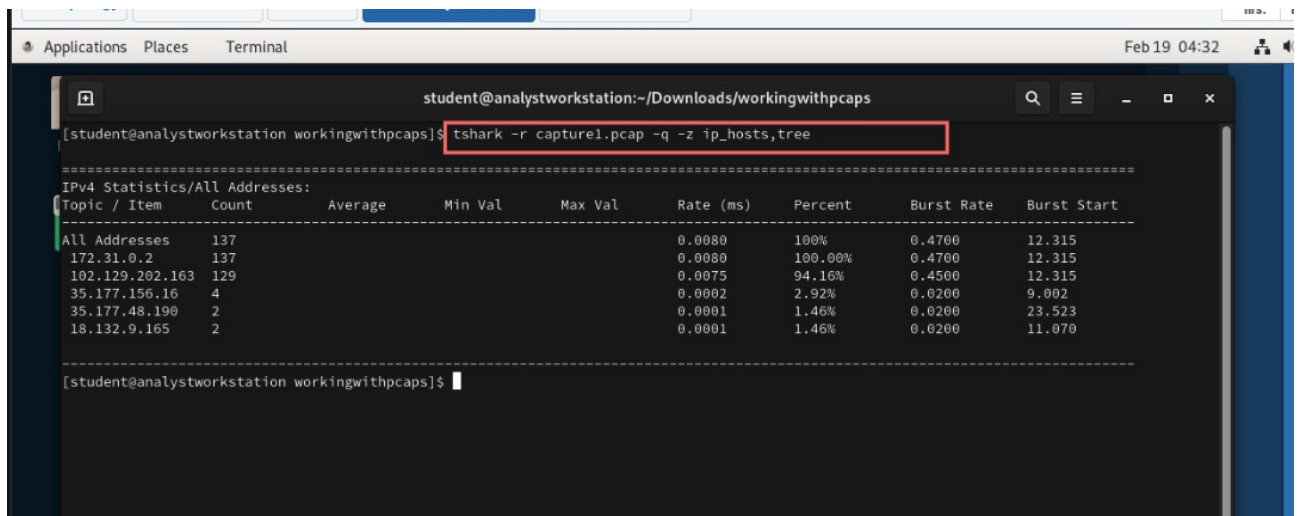


Figure 2.2. Host/IP statistics (*ip_hosts,tree*) for *capture1.pcap* highlighting the primary hosts.

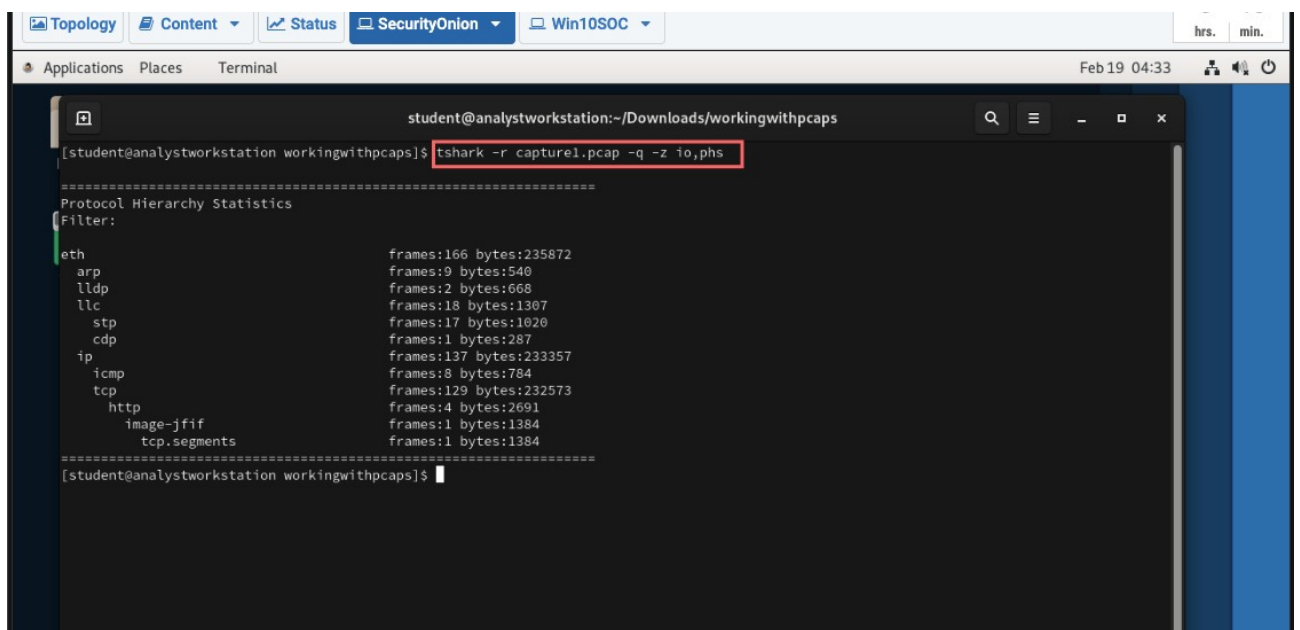


Figure 2.3. Protocol hierarchy statistics (*io,phs*) for *capture1.pcap* showing IP/TCP/HTTP and image-jfif.

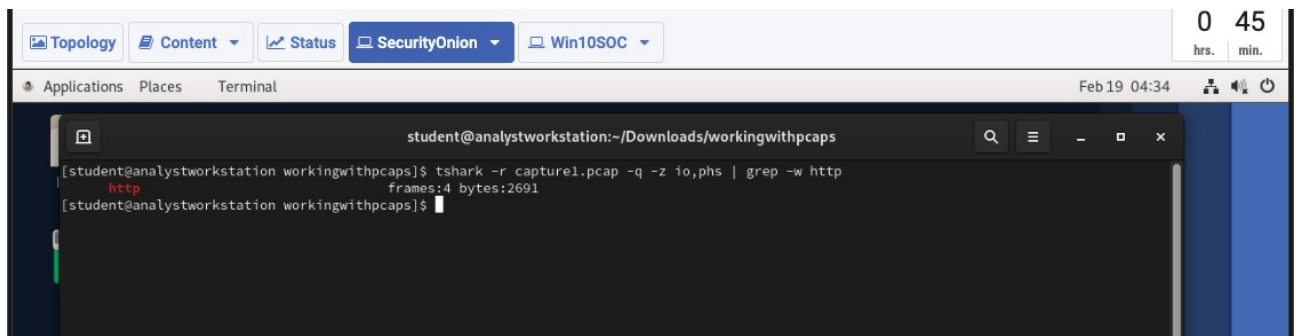


Figure 2.4. HTTP presence confirmed (frames count) by filtering protocol hierarchy output for HTTP.

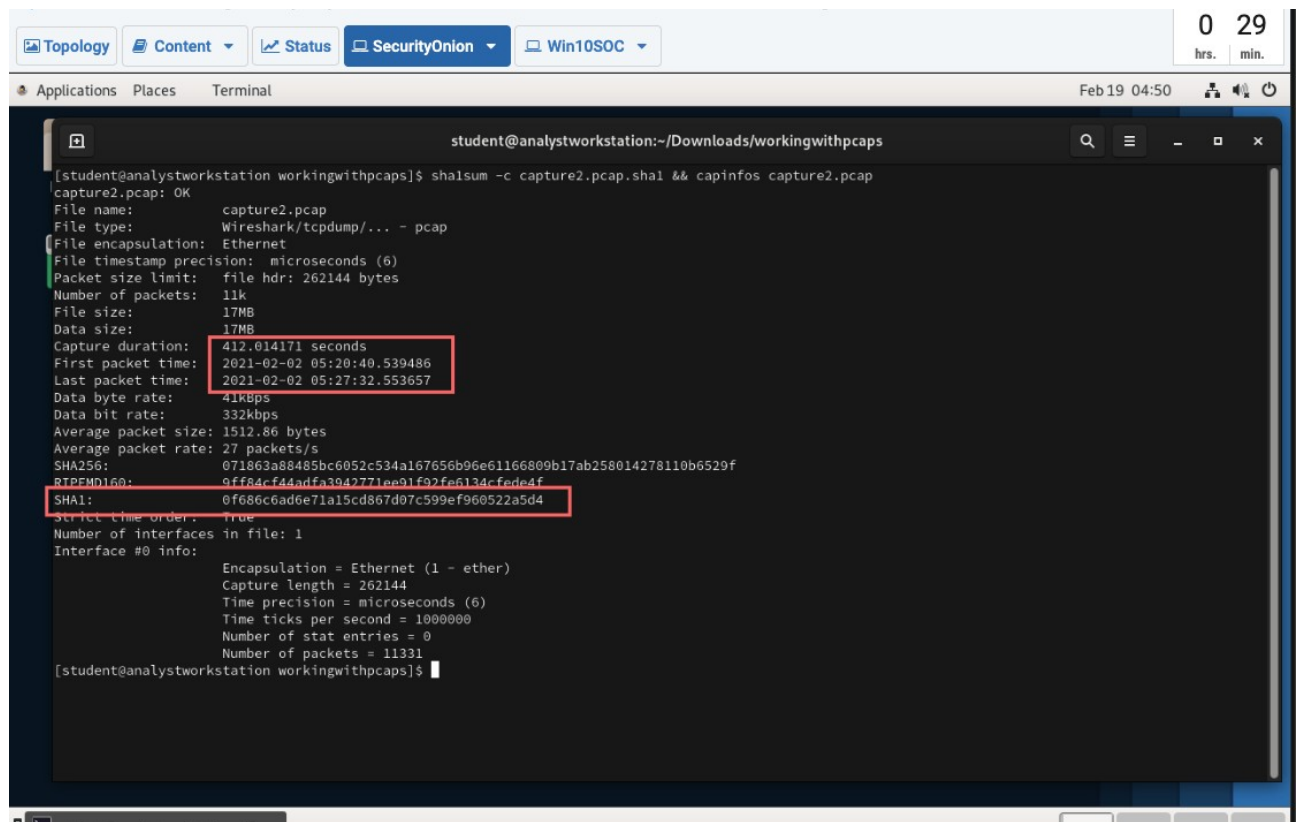
2.3 File Transfer Assessment (HTTP)

Based on the limited HTTP frames and the presence of image-jfif, this capture appears to contain a small amount of web browsing or content retrieval. No evidence of HTTP uploads is shown in the provided output.

3. Capture2.pcap Results

3.1 File Integrity and Metadata

SHA1 verification for capture2.pcap returned OK. capinfos indicates a longer capture window (~412 seconds) with 11,331 packets, allowing stronger conclusions about host behavior and repeated HTTP activity.



```
[student@analystworkstation workingwithpcaps]$ shasum -c capture2.pcap.shal && capinfos capture2.pcap
capture2.pcap: OK
File name: capture2.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 262144 bytes
Number of packets: 11k
File size: 17MB
Data size: 17MB
Capture duration: 412.014171 seconds
First packet time: 2021-02-02 05:20:40.539486
Last packet time: 2021-02-02 05:27:32.553657
Data byte rate: 41kBps
Data bit rate: 332kbps
Average packet size: 1512.86 bytes
Average packet rate: 27 packets/s
SHA256: 071863a88485bc6052c534a167656b96e61166809b17ab258014278110b6529f
RTPMD160: 9ff84cf44adfa3042771ee91f92fe6134cfede4f
SHA1: 0f686c6ad6e71a15cd867d07c599ef960522a5d4
Strict time order: true
Number of interfaces in file: 1
Interface #0 info:
  Encapsulation = Ethernet (1 - ether)
  Capture length = 262144
  Time precision = microseconds (6)
  Time ticks per second = 1000000
  Number of stat entries = 0
  Number of packets = 11331
[student@analystworkstation workingwithpcaps]$
```

Figure 3.1. SHA1 verification (OK) and capinfos metadata for capture2.pcap (packet count and time range).

3.2 Host and Protocol Identification

Host statistics show sustained communication between 172.31.0.2 and the external IP 193.176.211.47. Protocol hierarchy statistics confirm TCP-dominant traffic with minimal HTTP frames.

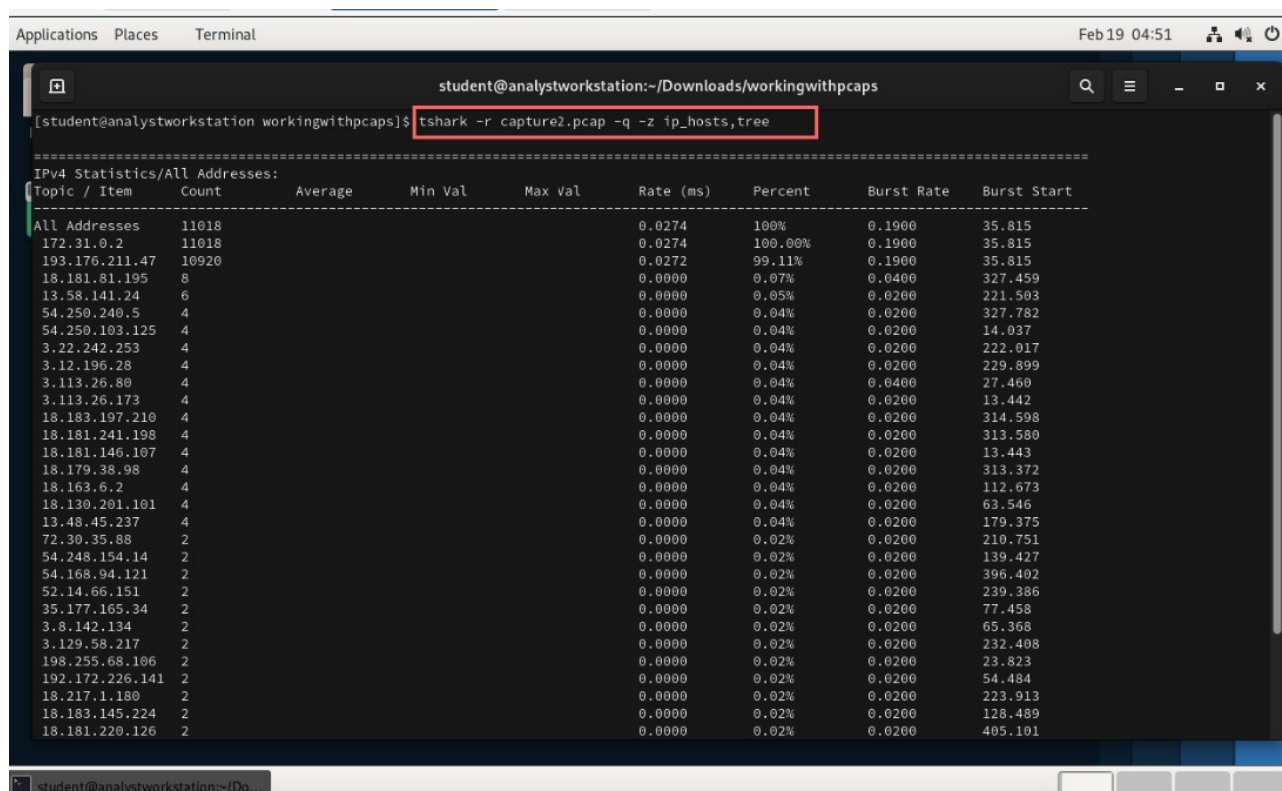


Figure 3.2. Host/IP statistics (ip_hosts,tree) for capture2.pcap highlighting the dominant peer hosts.

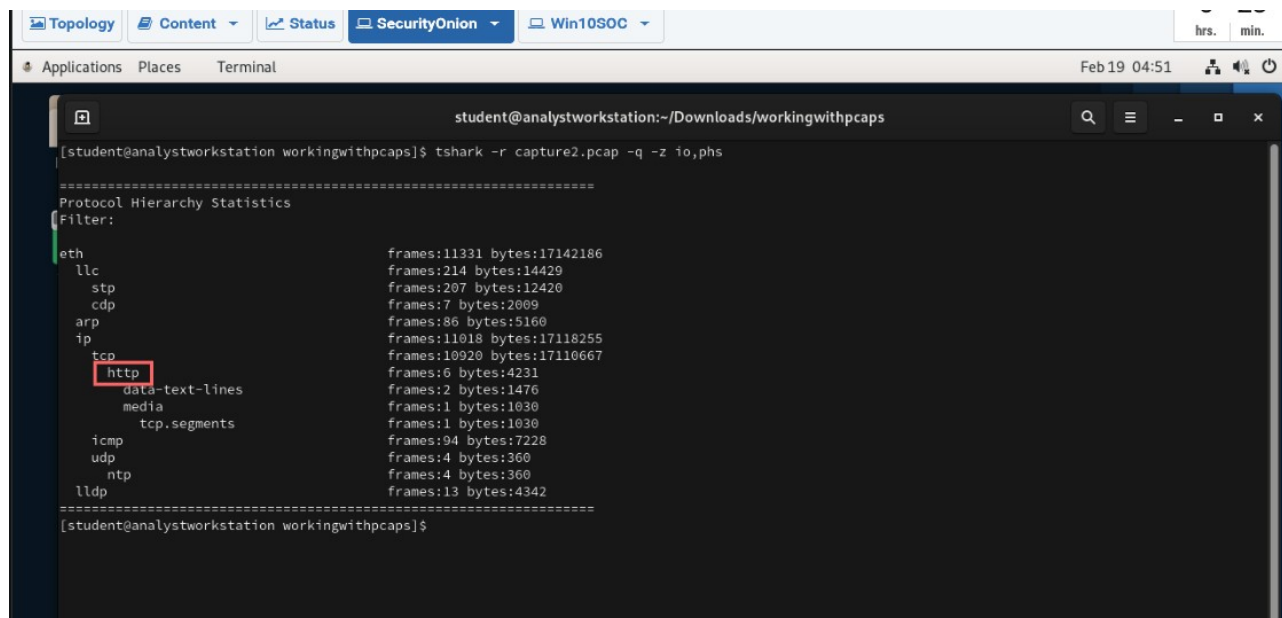


Figure 3.3. Protocol hierarchy statistics (io,phs) for capture2.pcap showing minimal HTTP traffic.

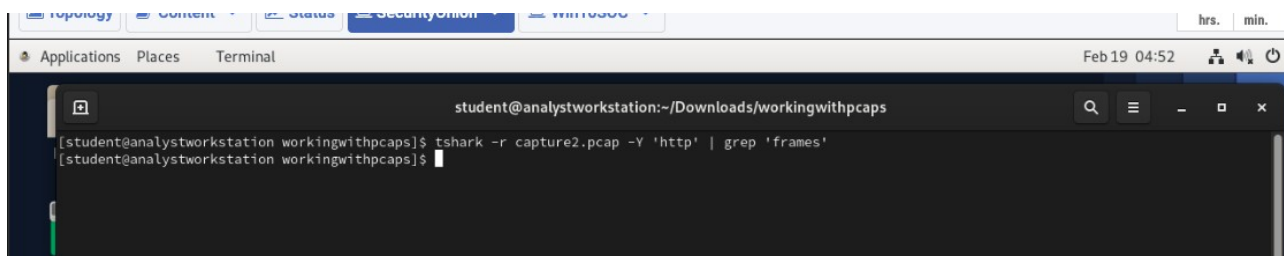


Figure 3.4. Direct HTTP filtering returned no additional frame summary output (limited HTTP activity).

3.3 HTTP Download Attempts (GET/POST)

HTTP traffic was extracted into a separate file for review. Multiple HTTP GET requests were observed for /thegoods/leetvid.zip. No POST requests were identified, indicating the activity was consistent with download attempts rather than uploads.

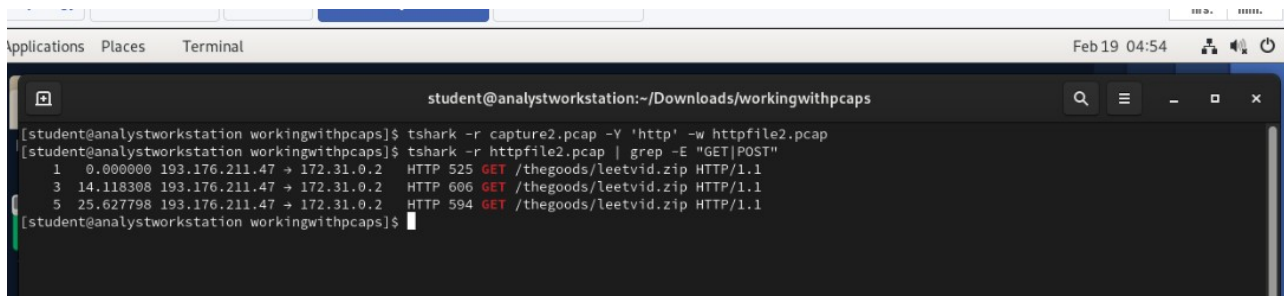


Figure 3.5. HTTP extraction and GET/POST review showing repeated GET requests for leetvid.zip and no POST.

3.4 HTTP Authentication Outcome

No http.authorization headers were identified in the extracted HTTP traffic. Additionally, the HTTP responses observed were 401 Unauthorized, indicating authentication failed and the requested resource was not successfully retrieved (no 200 OK responses were observed in the provided output).

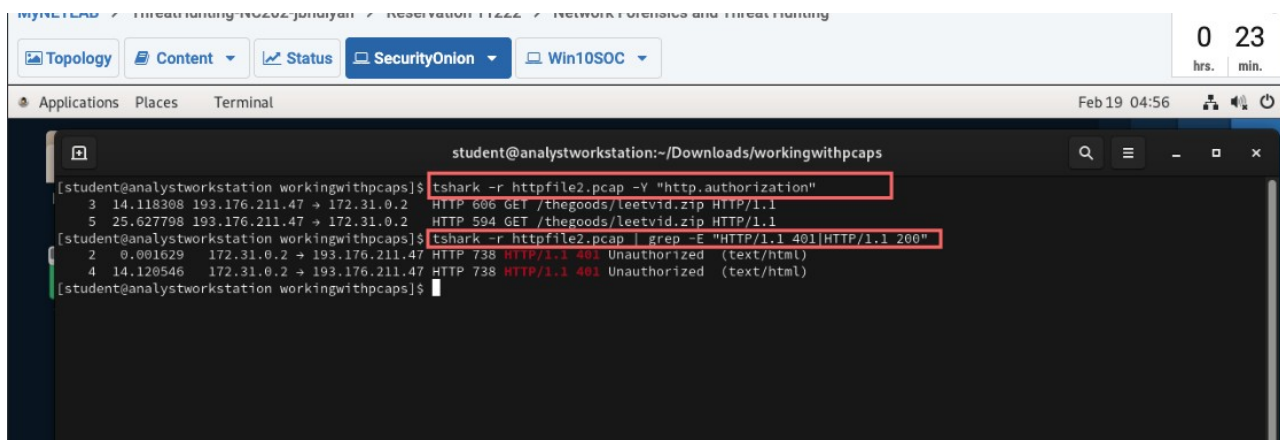


Figure 3.6. Authentication check shows no http.authorization and responses of HTTP/1.1 401 Unauthorized.

Conclusion

Across the analyzed captures, command-line packet inspection revealed clear indicators of protocol usage, authentication attempts, and file transfer behavior. ShellLab.pcap contained plaintext FTP authentication and a RETR request for xfil.rar, demonstrating the exposure created by unencrypted FTP. Capture1.pcap showed brief TCP/HTTP activity consistent with limited web content retrieval. Capture2.pcap contained repeated HTTP GET attempts for leetvid.zip, but response codes indicate authentication failures (401), supporting the conclusion that the download was not successfully completed.

Overall, the lab reinforced that rapidly scoping captures with capinfos and tshark statistics, then pivoting into protocol-specific filters, provides an efficient and defensible workflow for network forensics.

References

- Wireshark/tshark documentation (Wireshark User's Guide and tshark man page).
- capinfos man page (Wireshark suite).
- GNU coreutils shasum/sha1sum documentation.